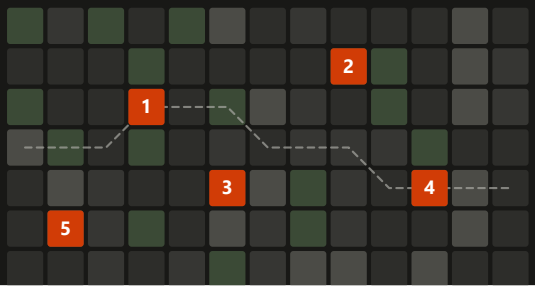
Canopia

Where should we plant first, and why?

An explainable screening tool, from free satellite and map data to a field-checked plan.

Pilot township: City of Elmina | Team Triple T
Sime Darby Property University Hackathon 2026, Planet track



SUPPORTING MATERIALS

A screening tool that ranks candidate tree-planting sites across a township from satellite measurements, explains every rank, and carries the shortlist through field checks to planting.

Live demo	https://canopia-app.vercel.app no login needed
Code	https://github.com/mpip0505/simedarbyhackathon
Team	Triple T, Universiti Malaya. Planet track, theme *Township of Tomorrow*
Pilot township	City of Elmina, chosen to test the method so it can be repeated in other townships

564
candidate 30 m sites ranked and explained

+51%
more exposed footpath when 10 sites are chosen as a set

14 of 20
top sites stay in the top 20 across 2,000 weightings

75
already-green sites flagged to protect first

> **Read this first.** This is a pilot of the method on real Elmina satellite and map data, not a pilot of real planting. The candidate sites are **simulated** (we do not have Sime Darby Property's site list), there is **no ground truth**, and no field walks or user tests have been run yet. Canopia ranks where to look first. It is not a tree count, a carbon figure or an approval to plant.

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2. THE PROBLEM, THE USER, WHAT IS DIFFERENT

Choosing where to plant first, with evidence

Sime Darby Property has a stated target of 50,000 endangered trees by 2030 and, in City of Elmina, a 5 km Urban Biodiversity Corridor linking six parks. Every season someone decides which sites get trees first. Heat on bare ground, gaps in the corridor and unshaded footpaths all matter, and without a repeatable method each choice is hard to explain or compare. Satellite maps do not settle it: independent maps put Elmina's tree cover anywhere from about 4% to 28%.

Who it is for

User	What they need	What Canopia gives them
Sustainability and landscape team (primary, a hypothesis to confirm)	decide each season's planting and justify it	a ranked, explained shortlist, adjustable priorities, field sheets and a Tracker to planting
Township planners and management	explain why a site was chosen	a reason per site, rank steadiness, a frozen record of the original shortlist
Residents and wildlife (beneficiaries)	shade on walking routes, habitat links	scores that target unshaded footpaths and corridor gaps

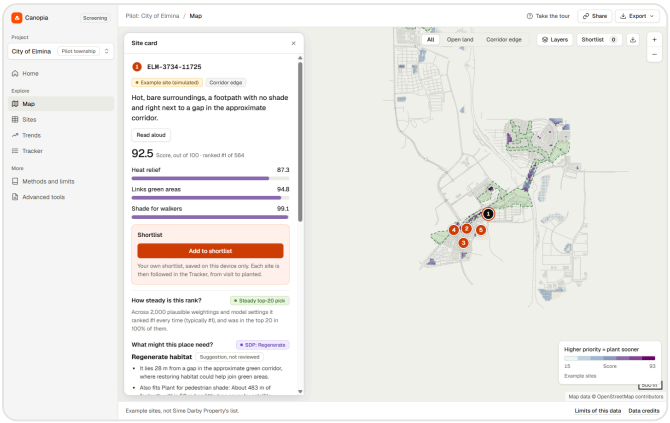
What makes Canopia different

Point	What it means, with a number you can check
Measured, not assumed	Every site score is computed from satellite measurements of the township: Sentinel-2 vegetation and Landsat ground heat (cloud-masked two-year medians), plus OpenStreetMap paths, parks and buildings. The sites are simulated; the measurements under them are real.
Township-wide, tied to the corridor	All 564 candidate sites across the township are ranked, and closing gaps in the Urban Biodiversity Corridor is one of the three scores.
The best set, not just top scores	Choosing 10 sites as a set reaches 51% more exposed footpath and about 3 times the corridor gap of the plain top 10, for a 6-point lower average score.
Says how sure it is	14 of the top 20 stay in the top 20 in at least 80% of 2,000 plausible weightings. The disagreement between independent canopy maps is shown on screen.
Protect before planting	75 sites are already green by satellite and are flagged for an ecologist before any planting, in Sime Darby Property's Avoid, Regenerate and Conserve terms (a screening suggestion, not a finding).
From ranking to the ground	A Tracker follows each site from shortlist to visit, decision and planting; blind field sheets and a validated import compare field results with the model. A real site list drops in with one command, tested to reproduce our scores exactly.
Built to be checked	One command rebuilds every number from free data in about 4 minutes. Automated tests run on every push, and the published test counts are generated from the code.

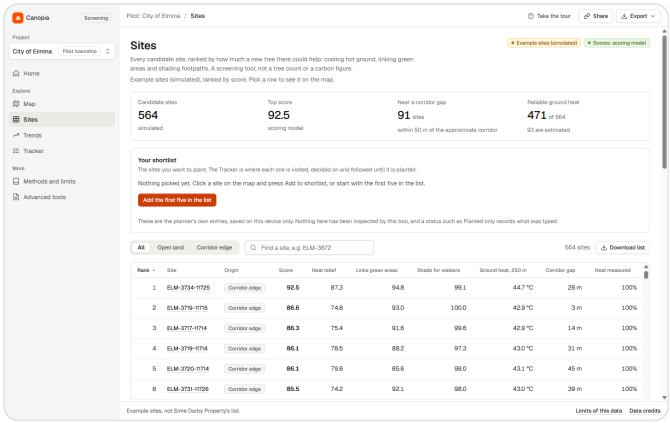
3. THE WORKING APP

Seven pages, live today

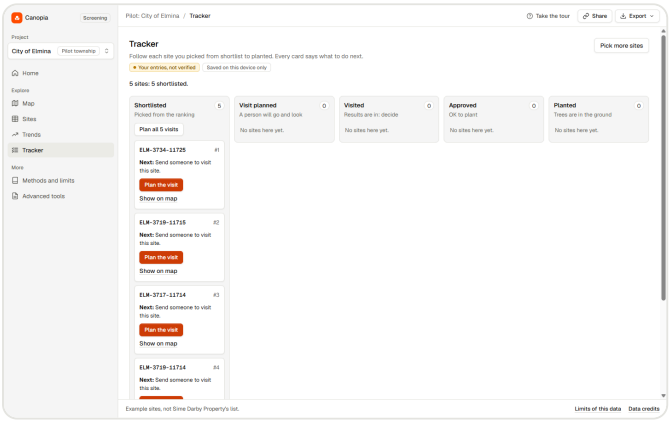
Home, Map, Sites, Trends, Tracker, Methods and limits, and Advanced tools. Everything a planner enters is their own record, saved in the browser; nothing is inspected or approved by the tool.



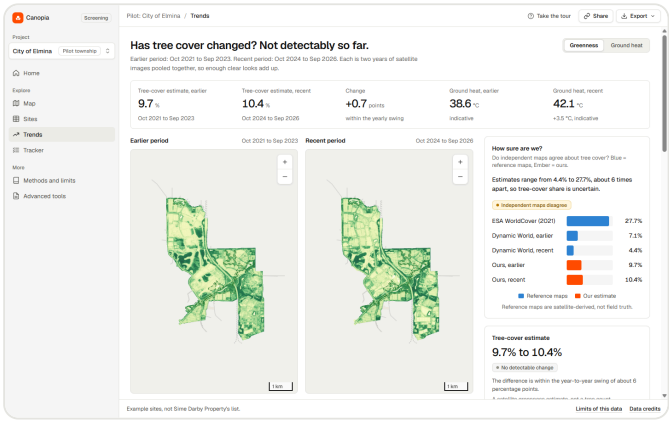
Map and site card. Site 1: "Hot, bare surroundings, a footpath with no shade and right next to a gap in the approximate corridor." Score 92.5; ranked first in every one of 2,000 weightings.



Sites. Every site in one sortable table, with its three scores, ground heat and distance to a corridor gap.



Tracker. Each picked site moves from Shortlisted to Visit planned, Visited, Approved and Planted, with one next step per card.



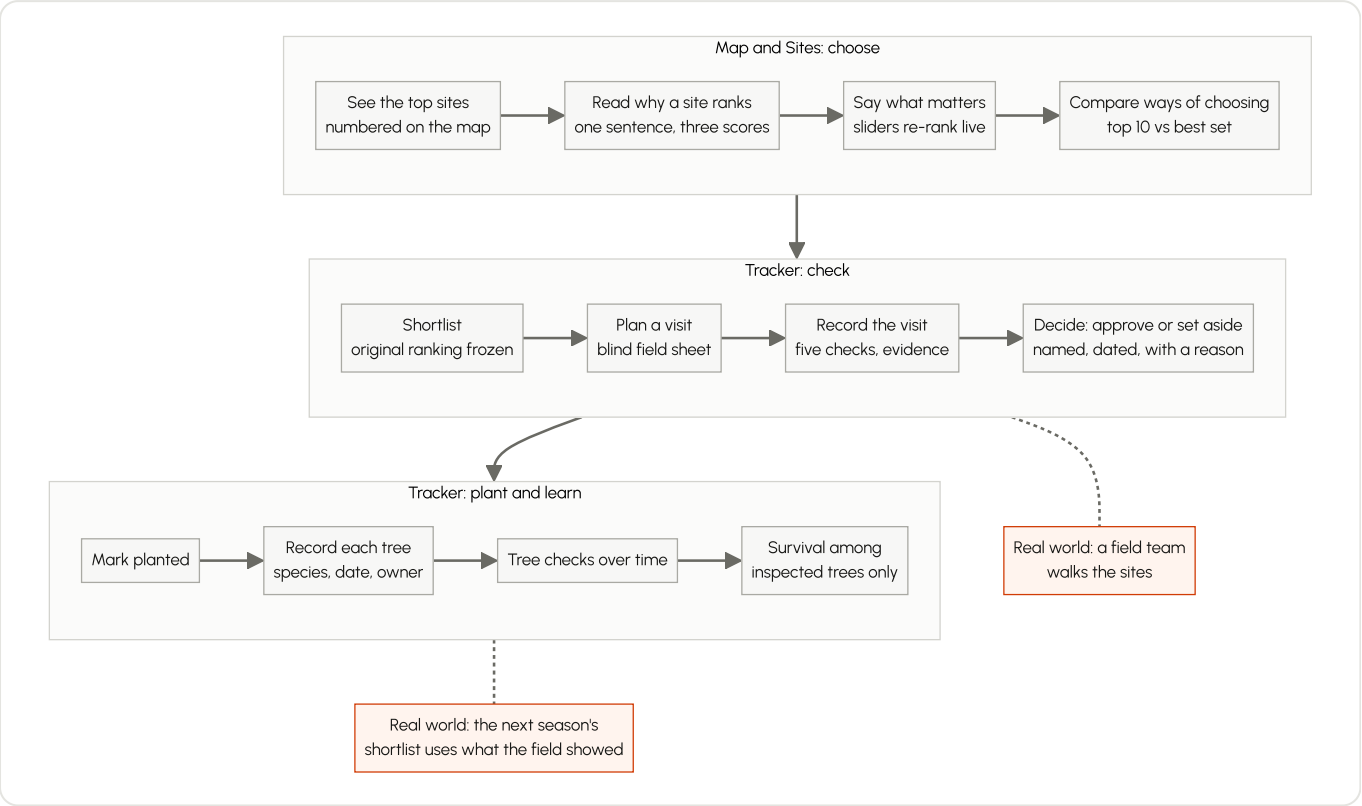
Trends. Tree-cover estimate 9.7% to 10.4%: within the yearly swing, so "no detectable change", beside the independent maps that disagree.

Screenshots of the deployed app, example sites (simulated), City of Elmina pilot.

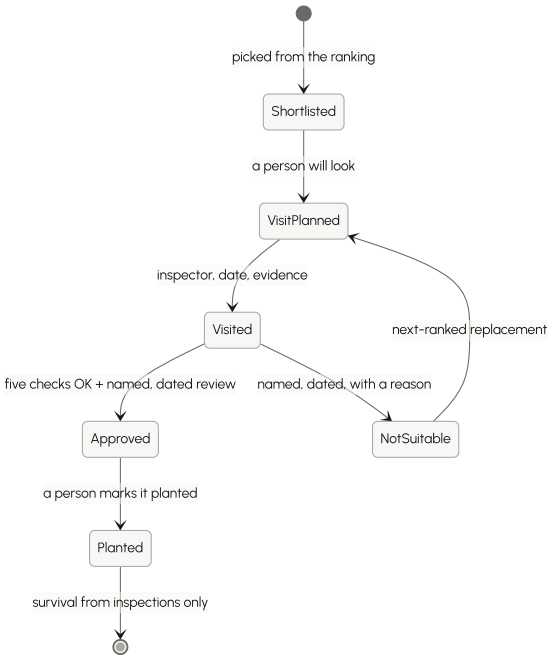
4. THE USER FLOW

From a ranked map to trees in the ground

The two shaded boxes are the real-world work a pilot with Sime Darby Property adds; everything else runs in the app today.



What a site goes through, and what the tool will not let a planner claim



The five field checks are land access, utilities, space and soil, maintenance access and existing trees. Each starts as Not checked; completing them is never an approval on its own.

5. HOW IT WORKS

A measured pipeline, served offline

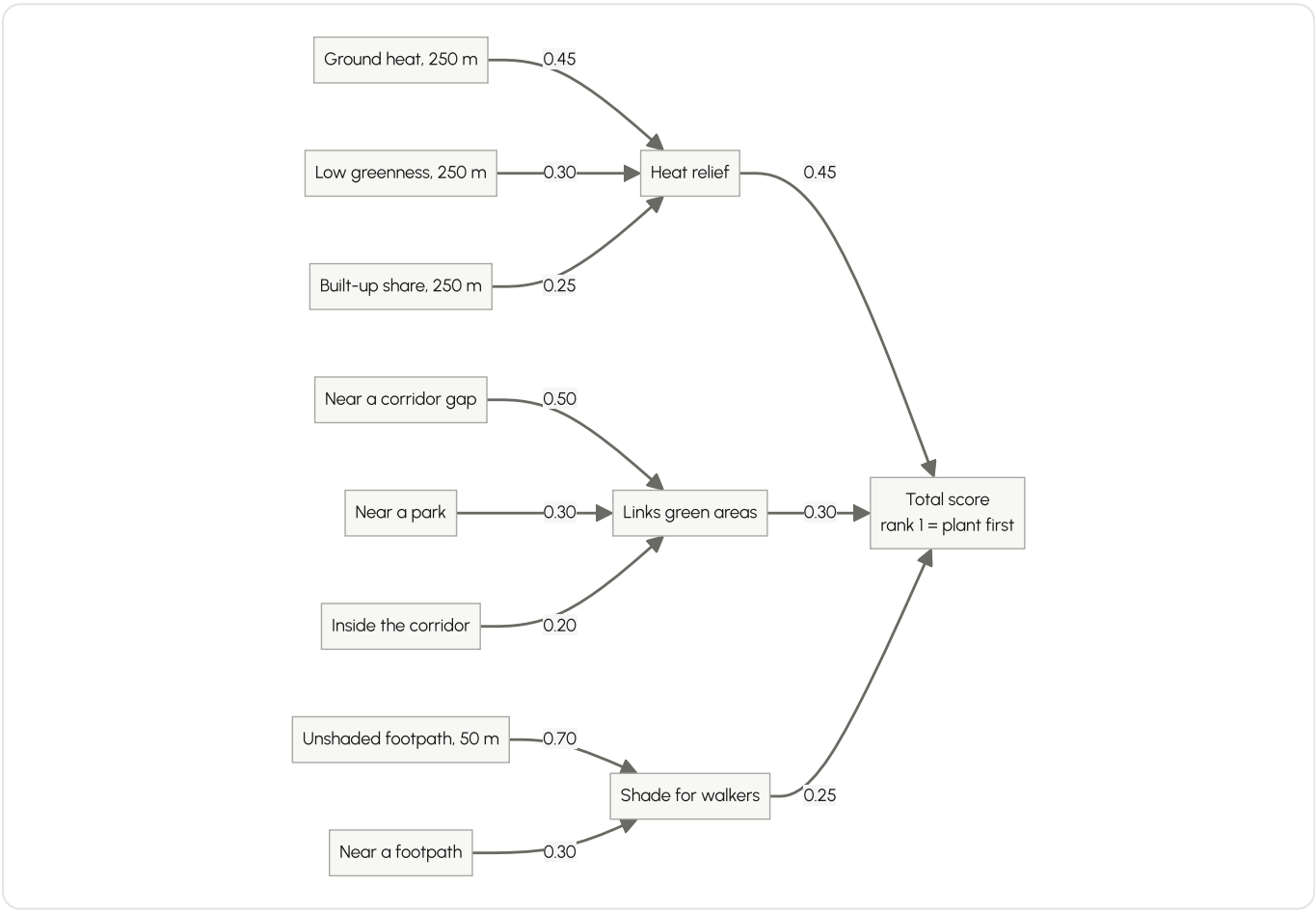
The build turns imagery and map data into versioned files; it needs Earth Engine and the network. The runtime serves those committed files with neither, so the demo works offline and on a read-only host.

Sentinel-2 greenness, Landsat ground heat, OpenStreetMap → Cloud-masked 2-year medians; approximate corridor → 564 candidate cells, about 30 features each → Scoring model and reports → Committed files, versioned by a data_version hash → FastAPI, read-only → Web app, no build step

Steps 1 to 5 run at build time (Earth Engine and the network). Steps 6 and 7 run anywhere with neither, including the public deployment.

How a site is scored

Three scores from eight named inputs, then a weighted total. There are no labels to fit to, so nothing is trained; every number traces to its inputs. The planner can change the three right-hand weights with sliders.



Heat, greenness and built-up share are percentile ranks within the township ("hotter than 90% of sites"), because 100 m thermal data supports ranking, not exact temperatures. Distances use a smooth decay. The heat term is pulled towards the middle where ground temperature had to be filled in from neighbouring pixels.

6. EVIDENCE

What we checked, and what we did not

We checked consistency and robustness thoroughly. We have no ground truth, so none of this shows the ranking is right; it shows it is consistent and stable.

Check	Result
Imagery go or no-go, before building anything	GO for both periods and both satellites: only 27 of 144 Sentinel-2 scenes in the baseline were under 40% cloud, yet every pixel had at least 3 clear looks.
Canopy threshold sensitivity	About 18%, 10% and 3% canopy at NDVI 0.5, 0.6 and 0.7, with a year-to-year swing of about 5 points. So canopy is used to rank sites, never quoted as a percentage.
Independent cross-check	ESA WorldCover about 28%, Google Dynamic World about 7% falling to 4%, ours about 10%. Best agreement at NDVI 0.6 (F1 0.51).
Rank steadiness	14 of the top 20 stay in the top 20 in at least 80% of 2,000 settings of the weights and judgement constants; rank correlation with the median ranking 0.999.
Method agreement	A heat-only ranking shares none of our top 20 sites; an equal-weights version shares 18 of 20. The weights matter, which is why the planner sets them.
Before and after	Our canopy proxy moved from 9.7% to 10.4%, inside its yearly swing: reported as "no detectable change", not as growth.
Reproducibility	A full rebuild took about 4 minutes and reproduced the committed sites; every build carries a data version and stale scores are flagged.
Your own site list	Scoring a simulated site's own centre point through the import reproduces its committed scores and rank exactly (automated test).
Software quality	Python, model and frontend unit tests run in CI on every push; a real-browser test covers every page, keyboard use, hostile data and an accessibility scan.

What we did not do

- **No ground truth:** no tree inventory, LiDAR, drone survey or field plots.
- **No real planting-site list:** the 564 sites are simulated 30 m cells on open land, and every screen says so.
- **No measured benefit:** reach numbers count what lies near the sites; they are not cooling, health, survival or carbon results.
- **No second township, no field walk, no user test yet.** The field kit is tested on synthetic answers only, and the usability protocol is written but not yet run.

7. IMPACT

What the first sites reach, and how a pilot would measure benefit

Today Canopia measures **reach** (what lies near the sites, each footpath point, home and gap counted once), not **benefit**. For the first 50 sites in the default ranking:

5,400 m
of footpath with little tree cover within 50 m (8% of the township's)

349 of 968
mapped homes within 250 m

70%
of the approximate corridor's gaps (about 1,150 of 1,650 m) within 50 m

Choosing sites as a set instead of by top score changes what a fixed budget reaches. For 10 sites, all three goals weighted equally:

Way to choose 10 sites	Footpath, little tree cover	Mapped homes	Corridor gap	Average score
Top 10 by score	2,120 m	242	350 m	86.2
Spread out (at most 3 per 250 m block)	2,280 m	242	410 m	85.6
Best coverage	**3,210 m (+51%)**	**262 (+8%)**	**1,030 m (+194%)**	80.0

The pilot measurement plan

What to measure	Baseline today	How a pilot would measure it	When
Footpath with little tree cover near planted sites	about 65,900 m township-wide; 5,400 m near the first 50 sites	re-run the count on new imagery; timed shade walks on the same routes	each season
Planted-tree survival	none	count live and dead trees at planted sites	6, 12, 24 months
Tree cover at planted sites	each site's canopy proxy	field or drone sample at planted and unplanted sites	12, 24 months
Does the ranking point at real shade gaps?	none	blind field sheet for top and lower ranked sites	before planting
Corridor gaps closed	about 1,650 m of gaps	re-count against the official corridor polygon	each season

No targets are set here, because nothing in our data supports a number; the developer's own targets are theirs to set against.

8. COMPARISON, COMMERCIAL CASE AND SCALE

Where Canopia sits, and how it repeats

Compared with published canopy prioritisers

Tool	Scale	Ranks by
Tree Equity Score (American Forests)	US census block group	canopy gap plus equally weighted social and heat indicators
i-Tree Landscape (USDA Forest Service)	US census block group	user-chosen indices and weights
Right Place, Right Tree (Boston, 2020)	census tract	heat vulnerability plus Landsat surface temperature, where planting is feasible
Canopia	**30 m candidate site**	**heat, corridor links and walkers' shade, with rank steadiness and a field loop**

A 2025 comparison of five tools in Austin found they agree on broad areas but differ on specific top sites, which is why Canopia shows how steady each pick is. We do not claim a new satellite method or to beat these tools.

Commercial case (hypotheses, to confirm with Sime Darby Property)

- **First customer:** the City of Elmina landscape operations team, with the sustainability team as reporting partner, through a bounded pilot: real sites, a sampled field inspection, and a comparison with the current shortlisting process.
- **Value multiplies** because the same pipeline and screens re-run for another township or question: more townships, corridor monitoring each season, where to inspect first after planting.
- **A real site list drops in today:** `python -m backend.my_sites` scores any list of points inside the study area with the same measurements and model, offline.
- **Another township** is estimated at a day or two of engineering (our estimate; not yet run elsewhere). What is Elmina-specific in the code is audited in the repository.
- **Costs** are low on data (free imagery and maps); the main costs are people and field checks.
- **Licensing, stated openly:** our Earth Engine registration is noncommercial. A runnable OpenStreetMap-only fallback exists; it is a proximity screen, not a second model, and whether the satellite workflow may be used commercially is a separate decision.

9. IMPLEMENTATION AND LIMITATIONS

From a working pilot to a season-by-season tool

When	What
Now	Prototype live on City of Elmina data with simulated sites; Tracker from shortlist to planted; field sheets ready.
Weeks 1 to 4	Score Sime Darby Property's real site list and swap in the official corridor polygon. The code already supports both.
Months 1 to 3	Walk about 30 sites with a blind field sheet to test whether the ranking reflects real conditions; an ecologist reviews the protect and regenerate suggestions.
Planting season	Plant from one season's ranking, re-run it each season, and compare before and after with survey data.
Next	Adapt to other Sime Darby Property townships.

What we need from Sime Darby Property: the planting-site list, the official corridor polygon, the current tree count, and one afternoon of field access.

Limitations

Limit	What it means
Simulated sites	30 m cells on OpenStreetMap open land, not a real planting list; OpenStreetMap under-maps parts of the township.
Approximate corridor	Built from OpenStreetMap parks: about 2 km of connecting spine against the stated 5 km. Connectivity results inherit this until the official polygon is supplied.
Coarse thermal data	Landsat surface temperature is about 100 m native; about 18% of the area is filled from neighbouring pixels for site scoring.
Canopy is a proxy	NDVI at or above 0.6, not a tree count; strongly threshold-dependent and noisy year to year.
Siting screen	Flags mapped roads, buildings and footpaths near a site; it is not a wildlife assessment.
Records are local	Tracker entries are saved in one browser; a shared, audited version is the first thing a real pilot would build.

Data credits

Map data (c) OpenStreetMap contributors (ODbL). Contains modified Copernicus Sentinel-2 data (European Space Agency). Landsat 8/9 Collection 2 Level-2 data courtesy of the U.S. Geological Survey. ESA WorldCover and Google Dynamic World used for the cross-check. Processed with Google Earth Engine (noncommercial use). Every figure in this document comes from a file in the repository and can be re-run.